

Digital Egypt Pioneers Initiative – DEPI

MindGuard-AI Documentation

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1. Project Proposal

1.1 Abstract

Depression is one of the most prevalent mental health disorders worldwide and can significantly affect an individual's emotions, behavior, and quality of life. With the widespread use of social media, users frequently express their thoughts and emotions through posts, images, and videos, creating digital footprints that may contain early indicators of psychological distress.

This project presents MindGuard AI, an AI-driven platform designed to support the early detection of depression by analyzing multimodal social media content. The system combines Natural Language Processing (NLP), Computer Vision (CV), and Deep Learning techniques to analyze textual and visual information, while incorporating a clinical assessment component to enhance prediction reliability. Based on the analysis, the platform provides explainable predictions and personalized recommendations that encourage users to seek professional support when needed.

MindGuard AI is intended as an AI-assisted screening tool rather than a diagnostic system, aiming to promote early awareness and intervention through an accessible web-based platform.

1.2 Introduction

Depression is among the most common mental health disorders worldwide and can have a profound impact on an individual's emotional well-being, daily activities, and overall quality of life. Early detection is essential for improving treatment outcomes; however, many cases remain unidentified until symptoms become severe.

The increasing popularity of social media has provided new opportunities for understanding users' emotional and psychological states. Through posts, comments, images, and videos, individuals often reveal thoughts and feelings that may reflect early signs of depression. Recent advances in Artificial Intelligence (AI), particularly in Deep Learning, Natural Language Processing (NLP), and Computer Vision (CV), have enabled the automated analysis of such multimodal data.

MindGuard AI is an AI-driven platform that leverages these technologies to analyze social media content and support the early detection of depression. By combining textual analysis, facial emotion recognition, and clinical assessment, the system generates

explainable predictions and personalized recommendations. The proposed solution is designed to assist in early mental health screening and encourage users to seek professional support when appropriate.

1.4 Problem Statement

Depression and other mental health disorders continue to increase worldwide, while many individuals do not seek professional help because of stigma, limited awareness, or financial constraints. As a result, early symptoms often go unnoticed until the condition becomes more severe.

Meanwhile, social media has become a major platform for expressing thoughts, emotions, and daily experiences. Research has shown that users' online activities can reveal behavioral patterns associated with depression; however, these indicators are rarely analyzed systematically. Existing assessment methods mainly rely on clinical evaluations and self-reporting, which may delay early detection. Therefore, there is a need for an intelligent system that can analyze multimodal social media content to support the early identification of depression indicators.

1.3 Project Objectives

The primary objective of this project is to design and implement **MindGuard AI**, an AI-based platform that supports the early detection of depression through the analysis of multimodal social media content and interactive user assessment.

The specific objectives of the project are:

- Developing a system that analyzes textual social media content using Natural Language Processing (NLP) techniques to identify linguistic and emotional patterns associated with depression.
- Implement a Computer Vision (CV) module to analyze facial expressions and visual cues that may indicate emotional distress.
- Develop a speech-based interaction module that enables users to communicate with the system through voice, providing speech recognition and conversational support for a more natural and accessible user experience.
- Integrate a PHQ-9 clinical questionnaire with AI predictions to improve assessment reliability.

- Provide explainable predictions using Explainable AI (SHAP) to increase transparency and user trust.
- Utilize advanced Deep Learning models for processing textual, visual, and speech data to enhance prediction performance.
- Generate explainable prediction results accompanied by personalized recommendations that encourage users to seek professional mental health support when appropriate.
- Design and implement an accessible web-based platform that provides AI-assisted mental health screening through an intuitive and user-friendly interface.

2. System Analysis & Requirements

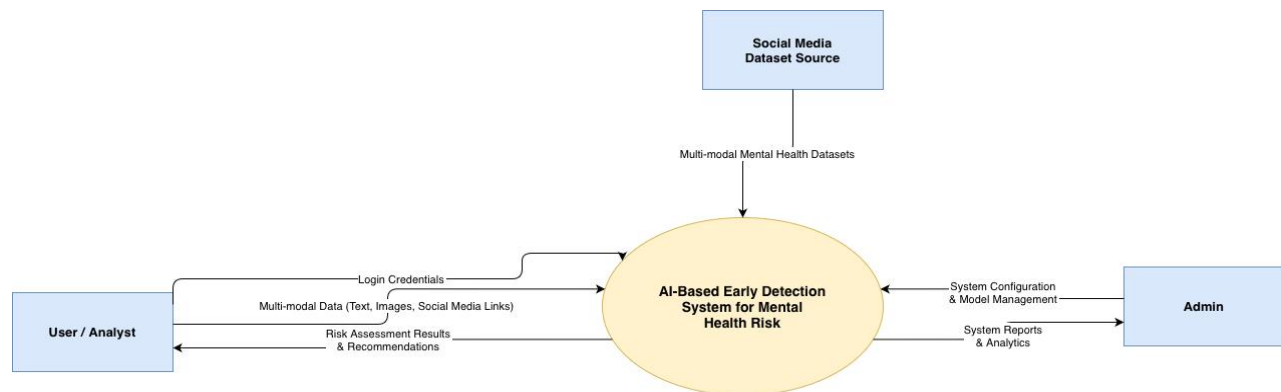
2.1 System Overview

MindGuard AI is a web-based platform that supports the early detection of depression through the analysis of user text, facial images, and voice interactions using Artificial Intelligence techniques such as Natural Language Processing (NLP) and Computer Vision (CV).

Users can create an account, log in securely, and maintain a personalized profile that stores their previous assessments and analysis history. The platform allows users to submit text, upload facial images, or interact with the system through voice. Based on the provided input, the AI model generates an initial depression risk prediction. This prediction is then combined with the user's **PHQ-9 clinical assessment** through a **decision fusion** module to produce a more reliable final depression risk probability. The system also provides **SHAP-based explanations** and personalized recommendations to help users understand the assessment results.

In addition to the prediction results, the system provides users with statistical insights such as mood trends, risk distribution, and recent analysis history, enabling them to monitor changes in their mental well-being over time. The platform is intended to support early mental health awareness and should not be considered a substitute for professional medical diagnosis.

Figure 1 illustrates the overall context of the proposed system and its interaction with external entities.



DFD Level 0 - Context Diagram
AI-Based Early Detection System for Mental Health Risk

Figure 1 DFD Level 0 – Context Diagram

The Context Diagram presents the system as a single process and illustrates its interaction with external entities. Users submit login credentials and multimodal data to the system and receive depression risk assessments and recommendations. The administrator manages system configuration and monitors reports, while social media datasets serve as the primary source for training the AI models.

2.2 Functional Requirements

User Management

- The system shall allow users to create an account and log in securely.
- The system shall maintain a personalized user profile containing previous assessments and analysis history.

Depression Risk Analysis

- The system shall allow users to submit textual content for analysis.
- The system shall allow users to upload facial images for emotion-based analysis.
- The system shall support speech interaction, enabling users to communicate with the AI through voice.
- The system shall analyze the provided input and generate a depression risk probability.

Explainable AI (XAI)

- The system shall provide explainable predictions using SHAP.
- The system shall highlight the textual or visual features that contributed most to the prediction.

Clinical Assessment

- The system shall provide the PHQ-9 questionnaire as part of the assessment process.
- The system shall record the questionnaire score for use in the final evaluation.

Decision Fusion

- The system shall combine the AI prediction with the PHQ-9 assessment to improve the reliability of the final result.

- The system shall generate personalized recommendations based on the combined assessment.

User Dashboard

- The system shall maintain a history of previous analyses.
- The system shall visualize mood trends over time.
- The system shall display the user's risk distribution.
- The system shall provide access to recent analysis results.

2.3 Non-Functional Requirements

Performance

- The system shall generate prediction results within an acceptable response time.
- The system shall efficiently process text, image, and voice inputs.

Security and Privacy

- User authentication and personal data shall be securely managed.
- The system shall protect user information and comply with applicable data privacy regulations.

Usability

- The platform shall provide an intuitive and user-friendly interface.
- The system shall be responsive and accessible from different devices.

Reliability

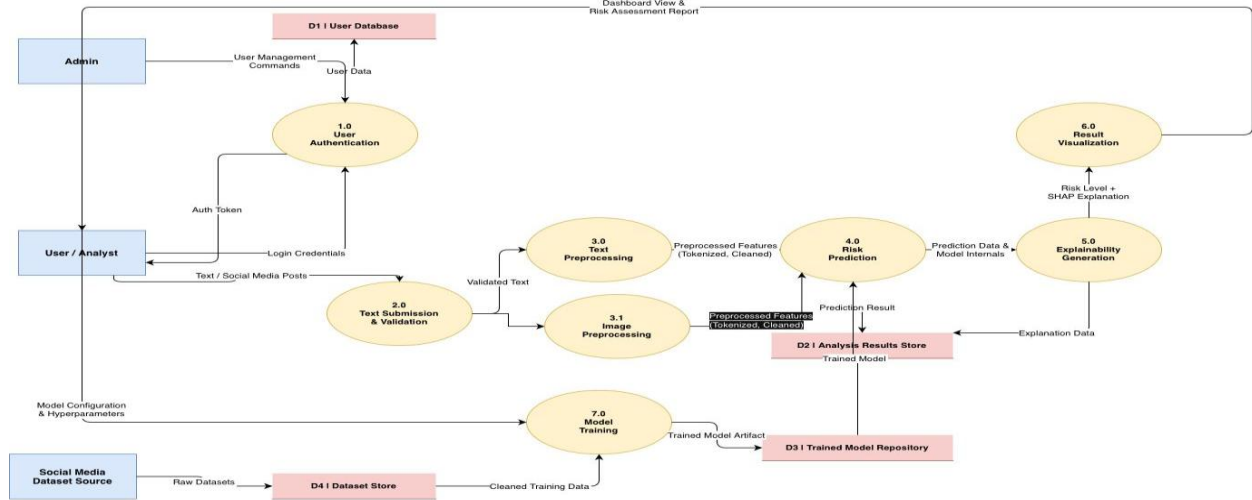
- The system shall provide stable and reliable operation with minimal downtime.
- The prediction process shall produce consistent results for similar inputs.

Explainability

- The system shall provide interpretable AI predictions using SHAP.
- The system shall clearly indicate that the results are intended for screening purposes and do not constitute a medical diagnosis.

2.4 Data Flow

To provide a more detailed view of the system, the Level 1 Data Flow Diagram decomposes the main process into its internal modules and illustrates how data flows between them.



DFD Level 1 - Detailed Data Flow Diagram
AI-Based Early Detection System for Mental Health Risk

Figure 2 DFD Level 1 – Detailed Data Flow Diagram

The Level 1 Data Flow Diagram illustrates the internal workflow of the proposed system. After user authentication, submitted text and images are validated and preprocessed before being passed to the prediction module. The generated results are explained through the Explainable AI module and presented to the user. The diagram also shows the interaction between the training process, dataset repository, trained model repository, and analysis results database.

3. System Implementation

3.1 System Architecture

The architecture of MindGuard AI consists of several integrated modules that work together to support early depression detection.

The main modules include:

- **AI-Based Analysis:** Processes user text and facial images using AI models.
- **Clinical Assessment:** Evaluates the user's responses to the PHQ-9 questionnaire.
- **Explainable AI:** Provides visual explanations of the prediction using SHAP and Grad-CAM.
- **Decision Fusion:** Combines AI predictions with the PHQ-9 assessment to generate the final depression risk.
- **User Dashboard:** Displays previous analyses, mood trends, and personalized recommendations.

3.2 Development Environment

The project was developed using the following technologies:

Component	Technology
Programming Language	Python
Development Environment	Jupyter Notebook
Backend Framework	FastAPI
Version Control	Git & GitHub
Deployment	Azure App Service

3.3 Computer Vision Module

The Computer Vision module analyzes users' facial expressions to identify emotional cues associated with depression.

The implementation includes:

- Face detection and alignment using **MTCNN**.
- Image preprocessing and data augmentation.
- Emotion recognition using a fine-tuned **EfficientNet-B4** model.

- Model optimization using **Focal Loss**, **AdamW**, and **Cosine Annealing**.
- Visual explanation of predictions using **Grad-CAM**.
- Model export in both **PyTorch (.pth)** and **ONNX** formats.

3.4 Natural Language Processing Module

The NLP module analyzes users' textual input to identify linguistic patterns associated with mental health conditions.

The implementation includes:

- Text preprocessing and cleaning using an appropriate preprocessing pipeline.
- Feature engineering for Machine Learning, Deep Learning, and Transformer-based models.
- Development and evaluation of multiple Machine Learning models, Deep Learning models, and the Microsoft **DeBERTa V3** Transformer model.
- Performance comparison to identify the best-performing model.
- Deployment of the selected Microsoft **DeBERTa V3** model using FastAPI for real-time text classification.

3.5 Explainable AI

To improve transparency, the system incorporates Explainable AI techniques.

- **SHAP** is used to explain text-based predictions by highlighting the most influential words.
- **Grad-CAM** visualizes the facial regions that contribute most to image-based predictions.

These explanations help users better understand the AI-generated results.

3.6 Clinical Assessment

The system includes the PHQ-9 questionnaire, a widely used depression screening tool.

The questionnaire score is combined with the AI analysis to improve the reliability of the final depression assessment.

3.7 Decision Fusion

The Decision Fusion module combines information from multiple sources, including:

- NLP prediction.
- Computer Vision prediction.
- PHQ-9 questionnaire score.

The combined information is used to generate the final depression risk probability and personalized recommendations.

3.8 Backend Implementation

The backend was implemented using FastAPI, providing RESTful API endpoints for communication between the AI models and the web application.

The API includes:

- /predict for image prediction.
- /health for checking system status.
- Interactive API documentation through Swagger UI.

3.9 Deployment

The Computer Vision API was deployed using Azure App Service.

The deployment includes:

- Automatic deployment from GitHub.
- Model storage using Git LFS.
- FastAPI running with Gunicorn and Uvicorn for production.

3.10 User Interface

The web application provides an intuitive interface that enables users to:

- Create an account and log in.
- Submit text, images, or voice input for analysis.
- Complete the PHQ-9 questionnaire.
- View depression risk results and recommendations.
- Monitor mood trends, risk distribution, and previous analyses through a personalized dashboard.

4. Conclusion

Mental health has become an increasingly important area of research, creating a growing need for accessible technologies that can support timely screening and promote greater awareness. This project introduced **MindGuard AI**, a web-based platform that combines multiple sources of information to provide a comprehensive mental health assessment while presenting the results in a clear and understandable manner.

Throughout the project, different machine learning and deep learning approaches were explored, evaluated, and integrated into a unified system capable of processing user inputs through an interactive interface. The platform also enables users to review previous assessments and monitor changes over time, providing a more informative experience than a single prediction.

Although the system is intended solely as a supportive screening tool and not a substitute for professional medical evaluation, it demonstrates how modern AI technologies can be applied to address real-world healthcare challenges. Future enhancements may focus on expanding supported data sources, improving prediction accuracy through additional datasets, and further strengthening system performance and usability for practical deployment.